CAB420 Machine Learning Project

Sports Image Classification Task

Group Number 91

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Summary

This project investigates which machine learning paradigm best balances accuracy and computational efficiency for 100-class sports image classification – a task made challenging by the visual similarity between many sport categories. The models chosen were a classical HOG + SVM pipeline, a deep learning CNN using EfficientNetB0 with transfer learning, and a Triplet Network using MobileNetV2 trained with triplet loss for metric learning. Each approach was evaluated on Top-1 accuracy, Top-5 accuracy, and macro F1-score to assess performance and consistency across classes. Training and inference time were recorded to compare computational efficiency.

The results demonstrate a clear performance hierarchy across the three approaches. The EfficientNetB0 CNN achieved the strongest overall results, attaining the highest scores across all evaluation metrics. This highlights that transfer learning with a modern CNN architecture is the most effective approach for this task. The Triplet Network performed competitively, particularly in Top-5 accuracy. However, its Top-1 accuracy fell short of the CNN, suggesting that its learned embeddings were not always discriminative enough to separate visually similar categories. Lastly, the HOG + SVM baseline achieved the weakest results across all metrics, highlighting the limitations of hand-crafted features for fine-grained visual classification at this scale.

In terms of computational efficiency, the HOG + SVM model, being the simplest model, trained significantly faster than both deep learning approaches. On the other hand, its evaluation time was the slowest of the three, reflecting the computational overhead of extracting HOG features. Both deep learning models required 60+ minutes to train; however, the CNN recorded the fastest evaluation time - benefitting from the inference efficiency of GPU – optimised deep learning frameworks. The most computationally expensive overall model was the Triplet Network, resulting from the additional complexity of triplet sampling and its two-stage training procedure.

These findings directly address the research question; deep learning approaches substantially outperform classical methods in classification accuracy. Amongst the three, transfer learning with the EfficientNetB0 CNN offered the best overall balance of performance and computational efficiency for this task.

1. Introduction and Motivation

Sports image classification presents an interesting challenge. While sports are visually distinct at a high level, many categories share common elements – similar body postures, overlapping equipment and shared venues – that make fine-grained discrimination non-trivial. Automated sports recognition has meaningful real-world applications in broadcasting media indexing, sports analytics, athlete monitoring programs, motivating the need for accurate and efficient classification systems.

The research question addressed by this project is: *Which machine learning paradigm – classical or deep learning – best balances accuracy and computational efficiency for 100-class sports image classification?* Therefore, to address this, three distinct machine learning approaches are evaluated and compared against an appropriate evaluation protocol.

This project builds upon a body of prior work in image classification, which has progressed from hand-crafted feature descriptors such as Histogram of Oriented Gradients (HOG) combined with Support Vector Machines (SVM) through to deep Convolutional Neural Networks (CNN) and metric learning approaches such as Triplet Networks. Comparisons between classical and deep learning approaches remain an active field of research, as deeper and more complex models do not yield improved performance across all problem domains. In this regard, a comparative evaluation of three distinct machine learning paradigms is undertaken, with the aim of gaining empirical insight into effectiveness, efficiency and suitability for each approach.

2. Related Work

Existing image classification approaches have developed from classical feature-based methods to deep learning models. Traditional methods often use hand-crafted descriptors such as Histogram of Oriented Gradients (HOG), which captures edge and gradient information from images [1]. When combined with a Support Vector Machine (SVM), this provides a simple and efficient classification pipeline. However, HOG features mainly describe low-level shape information, which can be limited for sports classification where categories may depend on equipment, athlete pose, background and playing environment.

CNN-based approaches improve on this by learning visual features directly from image pixels. Architectures such as ResNet, MobileNet and EfficientNet learn increasingly complex features across their layers, from simple edges to higher-level object and scene patterns. EfficientNet is particularly relevant because it improves the balance between accuracy and computational cost by scaling network depth, width and resolution together [4]. Transfer learning further supports this approach by using

ImageNet pre-trained weights, allowing models to reuse general image features learned from a large-scale dataset [5].

Existing sports image classification work has also used CNN-based models. For example, public EfficientNet-based implementations on the 100 Sports dataset report strong accuracy and F1 results [6], while SE-RES-CNN research applies residual connections and attention-based feature weighting to improve sports image classification performance [7]. These approaches show that CNNs are well suited to sports image classification, although more complex architectures may require greater computational resources.

Metric learning provides a different deep learning approach. FaceNet demonstrated the use of triplet loss to learn an embedding space where visually similar images are close together and different images are separated [2]. This idea can be adapted to sports classification using a Triplet Network. MobileNetV2 is commonly suitable for this type of approach because it provides efficient feature extraction with lower computational cost [3]. However, unlike CNN classifiers that directly output class probabilities, triplet-based methods usually require an additional matching step, such as nearest-centroid classification.

3. Data

3.1 Data Description

The 100 Sports Image Classification dataset was sourced from Kaggle and contains approximately 14,000 images across 100 diverse sports. Categories include conventional sports (e.g. tennis, basketball, football) as well as less common ones (e.g. wheelchair racing, sky diving, hydroplane racing), providing a dataset that is both visually diverse across categories and visually similar within certain sport clusters, making classification non-trivial.

The dataset is provided with a predefined split of 90 / 5 / 5:

- Training set: 13943 images (~139 per class)
- Validation set: 500 images (~5 per class)
- Test set: 500 images (~5 per class)

The images are provided as 224x244x3 jpg images.

3.2 Pre-processing

SVM

For the SVM model, class labels were encoded as integers using Label Encoder, as SVM classifier requires numerical targets. For HOG feature extractions, images were grey scaled as HOG operates on single-channel intensity images. HOG feature vectors were standardised using a Standard Scaler, transforming each feature to have a zero mean and unit variance, as kernel-based classifiers are sensitive to the scale of input features.

Triplet Network

For the Triplet Network, the original Kaggle split was replaced with a stratified 80/10/10 train, validation and test split to provide a more balanced evaluation across the 100 sport classes. This produced approximately 11,592 training images, 1,450 validation images and 1,450 test images after removing one unreadable image. All image paths were checked and missing or corrupted files were removed before training.

Images were loaded in RGB format and resized to 224 × 224 × 3 pixels to match the MobileNetV2 input size. The class labels were encoded as integers, while the original label names were retained for evaluation. MobileNetV2 preprocessing was applied inside the model to scale the input images into the expected format for the pre-trained backbone. During training, moderate data augmentation was applied, including random horizontal flipping, rotation, zoom and translation. This helped improve generalisation by exposing the model to variations in image position, scale and viewpoint.

CNN

For the CNN approach, the dataset was first reorganised to support a more reliable evaluation process. The original Kaggle split was recombined and then divided into a stratified 80/10/10 split for training, validation and testing. This resulted in 11,594 training images, 1,449 validation images and 1,450 test images. Stratification was used to ensure that the distribution of the 100 sport classes remained consistent across each split, reducing the risk of certain categories being under-represented during validation or testing.

Images were loaded in RGB format and resized to 224 × 224 × 3 pixels, which aligns with the expected input dimensions of EfficientNetB0. The labels were encoded as integer class values for model training, while the original sport names were retained for later interpretation of classification results.

No additional manual feature extraction was required for this method, as the CNN learns visual representations directly from the image pixels. This differs from the SVM approach, where HOG features needed to be extracted before classification. During training, light data augmentation was applied using random horizontal flipping, small rotations and zoom transformations. These augmentations were used to improve the model’s ability to generalise to variations in pose, camera angle and image framing.

3.3 Challenges

Upon class inspection, the training set is slightly imbalanced – where some classes have significantly less images than others. Sampled images had also exhibited varying lighting conditions and viewpoints. Additionally, many images contain multiple subjects, partial occlusions, and clustered backgrounds; contributing to challenges of intra-class variability and inter-class visual similarity.

3.4 Data Split

The initial validation and test sets contain 5 images per class, which may introduce high variance in evaluation metrics and limit the reliability of hyperparameters tuning. To address this, the dataset was repartitioned into an 80 / 10 / 10 split for training, validation and testing respectively.

4. Methodology

4.1 Support Vector Machine with HOG Features

The classical machine learning approach adopted combines a SVM classifier with HOG feature descriptors. Due to SVM's design property requiring input data to be represented as fixed-length numerical feature vectors - a feature extraction step is necessary prior to training. HOG was selected as it provides a robust and compact description of local shape and edge structure and has been widely validated as an effective descriptor for visual object recognition tasks [1].

HOG features were then extracted using the following configuration after pre-processes:

- Orientations: 9 bins, capturing gradient directions across 0 degrees to 180 degrees
- Pixels per cell: 16x16 pixels, defining the local region over which each gradient histogram is computed. This size chosen over a finer 8x8 alternative, for computational costs as the latter generates more HOG features and increases run time by roughly 4 times.
- Cells per block: 2x2 cells, defining the normalization block size
- Block normalization: L2-Hys, providing robustness to local illumination changes.

The SVM model was then trained using a Radial Basis Function (RBF) Kernel due to the nature of sports image data being non-linear. The regularisation parameter C was tuned via grid search over three candidate values $C \in \{0.1,\ 1.0,\ 10\}$. Due to training time constraints, the search space was intentionally kept narrow. The gamma parameter was set to scale allowing automatic adaption to feature scale. To enable fair comparison with deep learning models, probability was set to True, applying Platt scaling to convert SVM decision scores into calibrated class properties, from which Top-1 and Top-5 accuracies were derived from.

4.2 Convolutional Neural Network

The CNN method was implemented using EfficientNetB0 as a transfer learning model. This architecture was selected because it provides a strong balance between predictive performance and computational efficiency [4], which is suitable for a medium-sized image classification dataset with 100 output classes. Rather than training a CNN entirely from scratch, the model used weights pre-trained on ImageNet so that the lower-level and mid-level visual features learned from a large image dataset could be reused for the sports classification task [5].

The original EfficientNetB0 classification layer was removed and replaced with a task-specific classification head. The convolutional base was used to extract image features, while the new head mapped these features to the 100 sport categories. The added classification head consisted of a global average pooling layer, a dropout layer with a rate of 0.30, and a dense output layer with softmax activation. The softmax layer produced a probability distribution across all sport classes, allowing the class with the highest probability to be used as the Top-1 prediction.

The EfficientNetB0 base was initially frozen during training so that only the newly added classification layers were updated. This reduced the risk of overfitting and allowed the model to first adapt the final decision layers to the sports dataset. The model was trained using sparse categorical cross-entropy loss, as the class labels were represented as integer values. Adam was used as the optimiser with a learning rate of 1×10^{-3}.

Early stopping was applied by monitoring validation loss, allowing training to stop once the model no longer showed meaningful improvement. Model checkpointing was also used so that the best-performing model could be retained. The CNN was evaluated using Top-1 accuracy, Top-5 accuracy, macro F1-score, training time and test evaluation time, making its results directly comparable with the other approaches

4.3 Siamese Network with Triplet Loss

The Triplet Network was implemented as a deep metric learning approach. Instead of directly predicting class probabilities, the model learned an embedding space where images from the same sport class are placed close together, while images from different sport classes are pushed further apart. This was done using triplet loss, an approach commonly used in embedding-based recognition tasks such as FaceNet [2].

Each training sample consisted of three images: an anchor, a positive image from the same class, and a negative image from a different class. The model was trained to minimise the distance between the anchor and positive embeddings while increasing the distance between the anchor and negative embeddings.

MobileNetV2 was used as the shared convolutional backbone because it provides efficient feature extraction with relatively low computational cost [3]. The ImageNet pre-trained MobileNetV2 model was used without its original classification head. Its extracted features were passed through global average pooling, a dense layer with 256 units, batch normalisation, dropout of 0.30, and a final dense layer producing a 128-dimensional L2-normalised embedding.

The model was trained using triplet loss:

$$L = \max(d(a, p) - d(a, n) + margin, 0)$$

where d(a, p) is the distance between the anchor and positive embeddings, d(a, n) is the distance between the anchor and negative embeddings, and the margin was set to 0.3. This encourages images from the same class to be closer than images from different classes by at least the margin.

Training was completed in two stages. First, the MobileNetV2 backbone was frozen and only the new embedding layers were trained using Adam with a learning rate of 1×10^{-4} for up to 10 epochs. Then, the final 30 layers of MobileNetV2 were unfrozen and fine-tuned using a smaller learning rate of 1×10^{-5} for up to 5 epochs. Early stopping was used to reduce overfitting.

Since the Triplet Network outputs embeddings rather than class probabilities, classification was performed using a nearest-centroid approach. Training embeddings were averaged to create one centroid for each sport class. Test images were then classified by comparing their embeddings to all class centroids using cosine similarity. The closest centroid was used for Top-1 prediction, while the five closest centroids were used to calculate Top-5 accuracy. This allowed the Triplet Network to be evaluated using the same metrics as the other models: Top-1 accuracy, Top-5 accuracy, macro F1-score, confusion matrix, training time and evaluation time.

5. Evaluation and Discussion

5.1 Results Table

Models	Top 1 Accuracy / Accuracy	Top 5 Accuracy	Macro F1 Score	Training Time (s)	Embedding + Evaluation Time (s)
HOG+SVM	0.3069	0.5875	0.3015	1571.74	127.07
EfficientNetB0 CNN	0.9287	0.9896	0.9285	4227.04	24.95
Triplet Network	0.6841	0.9352	0.6815	4740.88	74.15

Table 1: Results table on test set performance for all approaches

5.2 SVM Evaluation

The SVM achieved the worst results across all metrics in this project; indicating that the model was struggling. This underperformance can be attributed to its reliance on hand-crafted HOG features, which lack the representational depth of learned features. When compared to CNN models, the SVM model cannot model hierarchical visual patterns, limiting their ability to distinguish between visually similar classes. Additionally, the high dimensionality of HOG descriptors and intra-class variation in the dataset likely made it difficult for the SVM to find a generalisable decision boundary.

To further investigate the SVM model's performance, it was also compared against other classical machine learning models in a Kaggle notebook evaluating the same dataset (adekgempi, 2026). F1 Scores are excluded from comparison as their models report weighted F1, making direct comparison misleading.

Model	Top-1 Accuracy	Top-5 Accuracy	Training Time (s)
HOG + SVM (Projects Model)	0.3069	0.5875	1571.74
Random Forest	0.2420	0.4900	8.4
XGBoost	0.2420	0.5060	50.6
LightGBM	0.2340	0.4840	51.1
Only SVM (RBF)	0.2220	0.5240	15.7

Table 2: Comparison table of Classical Machine Learning Models for image classification on this dataset

Another difference was their SVM model utilised PCA dimensionality reduction to manage computational complexity which was the opposite approach from our HOG pipeline. The primary trade-off was training time being significantly longer, which reflects the computational cost of kernel-based methods on high-dimensional feature spaces. When considering computational demands, the HOG+SVM also recorded the highest evaluation time despite being the simpler model compared to deep learning. This highlights a key limitation, as computing kernel distances against all support vectors during inference scales poorly as the dataset grows larger.

Despite its computational limitations, the HOG+SVM pipeline outperformed the other investigated classical machine learning models in previous studies. Which suggests that HOG descriptors provide a more discriminative representation for sports image classification than tabular statistical features. However, it still underachieves in comparison to the deep learning approaches which remain superior in both accuracy and efficiency.

5.3 CNN Evaluation

The EfficientNetB0 CNN achieved the best overall results in this project. On the test set, it reached a Top-1 accuracy of 0.9287, a Top-5 accuracy of 0.9896 and a macro F1-score of 0.9285. These results indicate that the model correctly classified most test images as its first prediction and almost always included the correct sport within its five highest-confidence predictions.

The macro F1-score is particularly important for this task because the dataset contains 100 sport categories. A high macro F1-score suggests that the model was not only performing well on a few easy categories, but was achieving strong performance across the class set more generally. This supports the suitability of EfficientNetB0 as a feature-learning model for sports image classification.

The CNN's strong performance can be explained by its ability to learn hierarchical visual representations directly from image data. Lower layers of the network can capture simple patterns such as edges, textures and colours, while deeper layers can capture more complex structures such as equipment, body positions, playing environments and sport-specific visual context. This gave the CNN a major advantage over the HOG + SVM method, which relied on manually extracted gradient features.

The CNN also performed better than the Triplet Network in Top-1 accuracy and macro F1-score. While the Triplet Network learned an embedding space based on visual similarity, the CNN was trained directly for multi-class classification using softmax outputs. This made the CNN better suited to selecting the exact sport category from 100 possible classes.

The CNN's results are also comparable with existing work on the same dataset. A public EfficientNetB3 implementation reported an accuracy of 92.72% and an F1-score of 91.76% [6]. The EfficientNetB0 model in this project achieved a similar or slightly stronger result, with 92.87% Top-1 accuracy and 92.85% macro F1-score. This is a positive outcome because EfficientNetB0 is a smaller model than EfficientNetB3. A published SE-RES-CNN model has reported stronger performance of approximately 98% accuracy on a 500-image test set [7]. However, this comparison should be interpreted carefully because the architecture, test split and training procedure are different from the 80/10/10 evaluation protocol used in this project.

The main limitation of the CNN was training cost. It required 4227.04 seconds, or approximately 70 minutes, to train (Figure1). This was significantly longer than what would usually be expected from a classical feature-based model. However, its final evaluation time was only 24.95 seconds, showing that the model was efficient once trained. Therefore, the CNN offers the best trade-off when predictive accuracy is the main priority, while its main disadvantage is the initial training time.

Some classes still showed weaker results. For example, figure skating pairs, shot put, sky surfing and golf had lower F1-scores than many other categories. These weaker results are likely caused by visual overlap with related classes. Figure skating pairs can be visually similar to figure skating men and figure skating women because they share similar rink backgrounds, clothing and poses (Figure2). Shot put can also resemble other athletics events, while golf may share outdoor environments with other field-based sports. This suggests that the remaining errors are mostly due to fine-grained visual similarity rather than a complete failure to learn useful features.

Overall, the EfficientNetB0 CNN was the most effective approach in this project. It achieved the strongest classification results, the highest Top-5 accuracy and the best macro F1-score. Its performance shows that transfer learning with a modern CNN is highly suitable for the 100 Sports Image Classification dataset.

5.4 Triplet Network Evaluation

The Triplet Network was evaluated on the re-split 80/10/10 dataset, using 11,592 training images, 1,450 validation images and 1,450 test images across 100 sport classes. The model achieved a Top-1 accuracy of 0.6841 and a macro F1-score of 0.6815. This indicates that the model correctly classified approximately 68% of test images when only the closest class centroid was considered. The similar values for accuracy and macro F1-score suggest that the model performed reasonably consistently across the 100 classes, rather than only performing well on a small number of dominant classes.

The model achieved a much higher Top-5 accuracy of 0.9352. This shows that although the closest predicted class was not always correct, the correct sport category was usually within the five most similar class centroids. This is an important result for sports image classification, as some sport categories share similar visual features, such as similar playing fields, equipment, uniforms or body poses. Therefore, the high Top-5 accuracy suggests that the embedding space learned by the Triplet Network was effective at grouping visually related sports, even when it could not always separate the exact class.

The Triplet Network required a total training time of 4,740.88 seconds, which is approximately 79 minutes. This made it computationally more expensive than simpler classical approaches, mainly because the method uses a deep MobileNetV2 backbone and trains on triplets of anchor, positive and negative images. The embedding extraction and evaluation stage took 74.15 seconds, which is relatively efficient once the model had been trained. This suggests that the main computational cost of the method is during training rather than inference.

Overall, the Triplet Network showed strong performance as an embedding-based classification method. Its main strength was its high Top-5 accuracy, showing that it learned meaningful visual similarity relationships between sports classes. However, its Top-1 accuracy was lower than its Top-5 accuracy, suggesting that some classes remained difficult to distinguish when they were visually similar. Future improvements could include using harder negative mining, longer fine-tuning, a larger embedding size, or a stronger backbone such as EfficientNet or ResNet to improve separation between closely related sport categories.

Conclusion

This project compared three approaches for 100-class sports image classification: HOG + SVM, EfficientNetB0 CNN, and a Triplet Network. The experiments met the project objective by showing how classical machine learning, direct CNN classification, and similarity-based learning differ in accuracy and computational efficiency.

The HOG + SVM model provided a useful baseline, but performed the weakest because hand-crafted HOG features could not capture the complex visual differences between many sports. The Triplet Network performed better and achieved strong Top-5 accuracy, showing that it learned useful visual similarity between classes. However, it was less effective for exact Top-1 classification because it required nearest-centroid matching rather than direct class prediction.

The EfficientNetB0 CNN was the best-performing model overall, achieving the highest Top-1 accuracy, Top-5 accuracy and macro F1-score. This shows that transfer learning with a modern CNN is the most suitable approach for this dataset. Although it required longer training time, it produced the strongest results and had efficient evaluation once trained.

Future work could focus on improving performance for visually similar classes, such as figure skating pairs, shot put, sky surfing and golf. For the CNN, this could involve fine-tuning later EfficientNetB0 layers with a smaller learning rate, allowing the model to adapt more specifically to sport-related features. Stronger data augmentation could also be tested, including brightness changes, cropping, translation and contrast adjustment, to improve robustness to different camera angles, lighting conditions and backgrounds.

For the Triplet Network, future improvements could include hard negative mining, where the model is trained on more difficult negative examples that are visually similar to the anchor image. This may help improve class separation in the embedding space. A stronger backbone, such as EfficientNet or ResNet, could also be tested to improve feature extraction.

References

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Appendix

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EfficientNetB0 Sports Classification Results
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Train size: 11594
Validation size: 1449
Test size: 1450

Training time: 4227.040916442871 seconds
Evaluation time: 24.947490215301514 seconds

Test loss: 0.2429443746805191
Test Top-1 Accuracy: 0.928719699382782
Test Top-5 Accuracy: 0.9896193742752075
Macro F1: 0.9285275348713129
```

Figure 1: CNN results

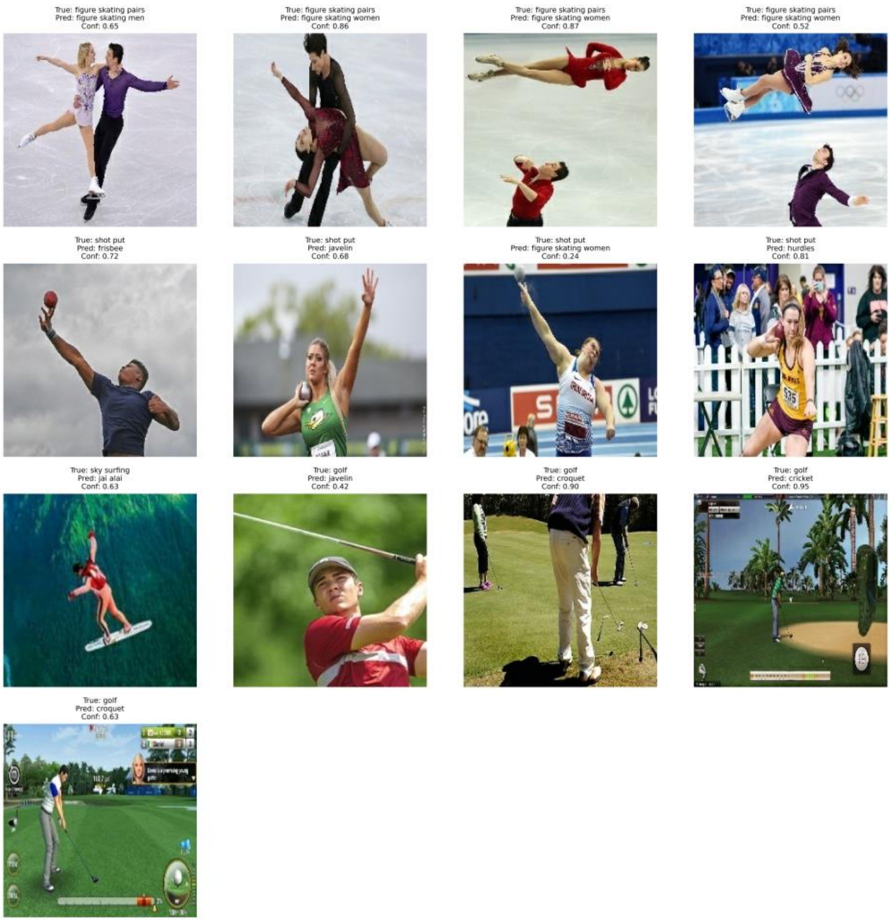

Figure 2: CNN worst class incorrect prediction

Contributions Table

Name	Contribution	Percentage
Brendan	Responsible for Introduction, Data and SVM models written components. Implemented the SVM model, train and evaluation.	33.3%
Fan-Bo	Responsible for the Triplet Network component. Implemented the Siamese-style Triplet Network with triplet loss using a MobileNetV2 backbone,	33.3%

	including triplet sampling, model training, fine-tuning and evaluation. Also contributed the Triplet Network preprocessing, methodology, evaluation discussion, result interpretation, references and relevant conclusion/future work content.	
Chen-Hsiang	Responsible for the CNN component. Prepared the stratified 80/10/10 dataset split, implemented the EfficientNetB0 transfer learning model, trained and evaluated the CNN. Also contributed the CNN preprocessing, methodology, evaluation discussion, appendix results, and relevant conclusion/future work content.	33.3%

Signatures

Brendan:

Chen-Hsiang:

Fan-Bo: